LEDiff: Latent Exposure Diffusion for HDR Generation

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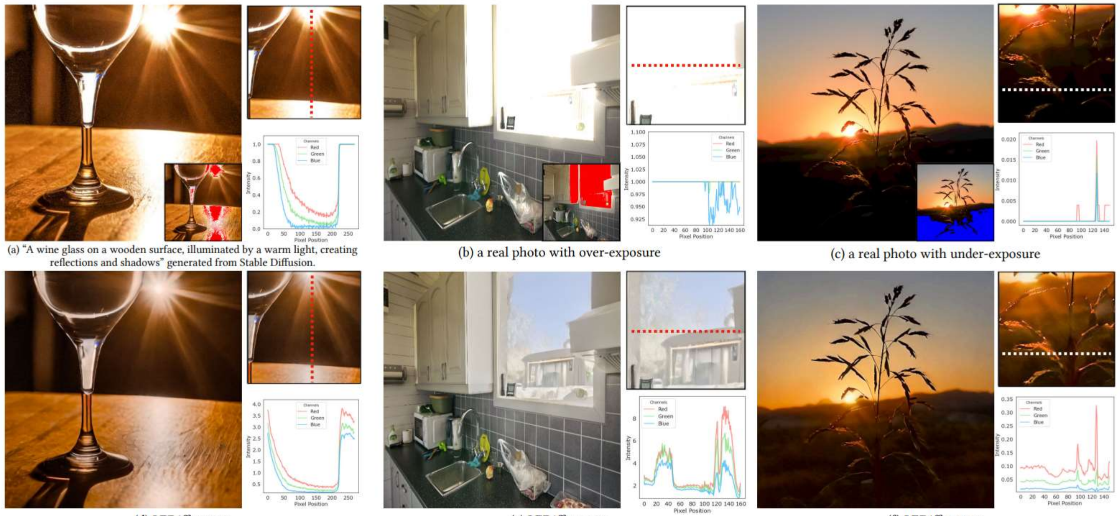


(a) "A wine glass on a wooden surface, illuminated by a warm light, creating reflections and shadows" generated from Stable Diffusion. (b) a real photo with over-exposure (c) a real photo with under-exposure

(d) LEDiff output (e) LEDiff output (f) LEDiff output

Figure 1. LEDiff enables high dynamic range (HDR) content generation with photorealistic details in both over- and under-exposed regions by performing exposure fusion in latent space, making it applicable to generated content and real photos mapped to the latent space. While existing generative models (e.g., Stable Diffusion) are restricted to low dynamic range (a) and standard cameras struggle to capture full scene dynamic range, causing clipping in highlights (b) and shadows (c), LEDiff restores both detail and dynamic range (d)–(f), as shown in scanline plots. All HDR images are tone-mapped for visualization and are best viewed on an HDR display. See the supplemental for more details.

- CVPR 2025
- Paper: https://openaccess.thecvf.com/content/CVPR2025/html/Wang_LEDiff_Latent_Exposure_Diffusion_for_HDR_Generation_CVPR_2025_paper.html
- Github: https://github.com/Hans1984/LEDiff

Backgrounds

Dynamic Range (DR)

- Definition
    - **maximum / minimum** measurable value
    - dimensionless quantity
    - In Imaging,
        - the ratio of two luminance values (cd/m^2)
            - https://en.wikipedia.org/wiki/Candela_per_square_metre
    - In Audio
        - the ratio of the amplitude of the ~
    - In Electronics..
        - the ratio of a maximum level of a parameter, such as power, current, voltage or frequency, to the minimum detectable value of that parameter.
    - ....

- https://pergolalights.net/what-are-lumens-lux-nits-and-candela/

DR depends on “what system”

- Scene Dynamic Range
  - brightest / darkest luminance in a scene
  - 장면 자체의 물리적 특성
- Camera Dynamic Range
  - the ratio of saturation to noise
  - sensor saturation / noise floor
  - 최대값: sensor saturation 직전
  - 최소값: noise보다 유의미하게 구분 가능한 최소 signal
  - SNR 기반 DR 정의와 동일한 맥락
- Display Dynamic Range
  - max brightness / min brightness
  - = contrast ratio (예: 1000:1)

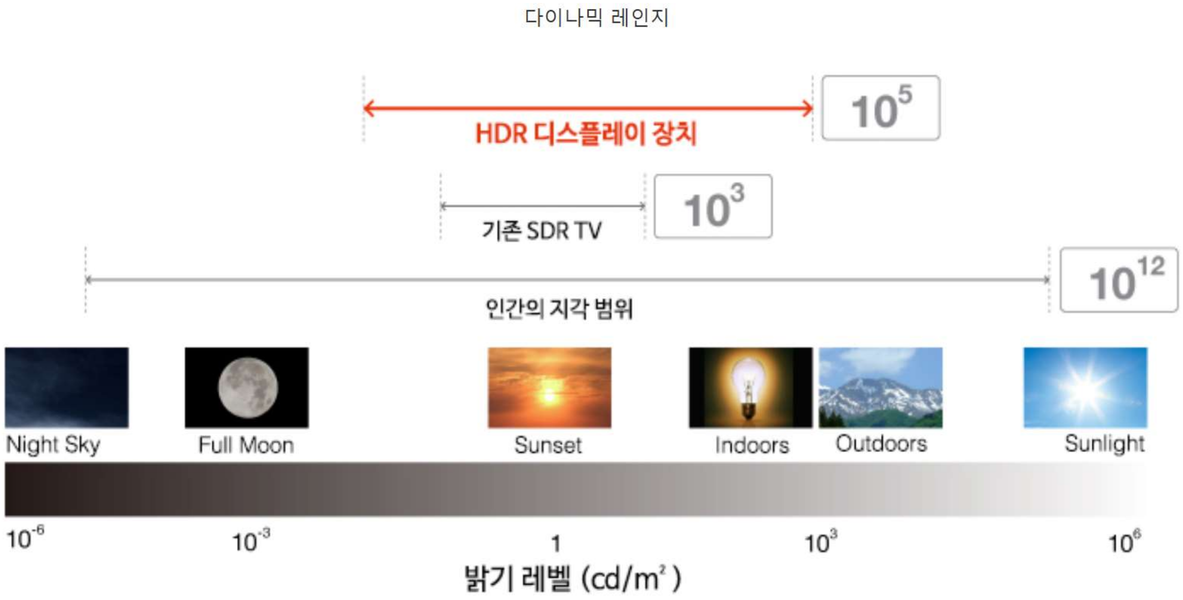
name	formula	example	context
contrast ratio	$\mathrm{CR} = (Y_{peak}/Y_{noise}) : 1$	500:1	displays
log exposure range	$\mathrm{D} = \log_{10}(Y_{peak}) - \log_{10}(Y_{noise})$ $\mathrm{L} = \log_2(Y_{peak}) - \log_2(Y_{noise})$	2.7 orders 9 stops	HDR imaging, photography
peak signal to noise ratio	$\mathrm{PSNR} = 20 \cdot \log_{10}(Y_{peak}/Y_{noise})$	53 [dB]	digital cameras

Table 1: Measures of dynamic range and their context of application. The example column illustrates the same dynamic range expressed in different units (adapted from [108]).

- https://pergolalights.net/what-are-lumens-lux-nits-and-candela/

Key mismatch

- Scene DR > Camera DR > Display DR
  - 밝은 영역 → saturation (clipping)
  - 어두운 영역 → noise에 묻힘



- https://www.cgkorea.co.kr/global/library/The_Ins_and_Outs_of_HDR_What_is_HDR.html

LDR & HDR

- **Low Dynamic Rage (LDR)**
  - 정의
    - 제한된 dynamic range를 가지는 이미지
    - 픽셀 값이 luminance와 비선형 관계
  - 핵심 특징
    - gamma correction이 적용된 값
    - pixel value ≠ 물리적 밝기
    - → luma 로 해석
  - 의미
    - 디스플레이 / 저장에 적합한 형태
    - 사람 눈에 보기 좋게 인코딩 된 값

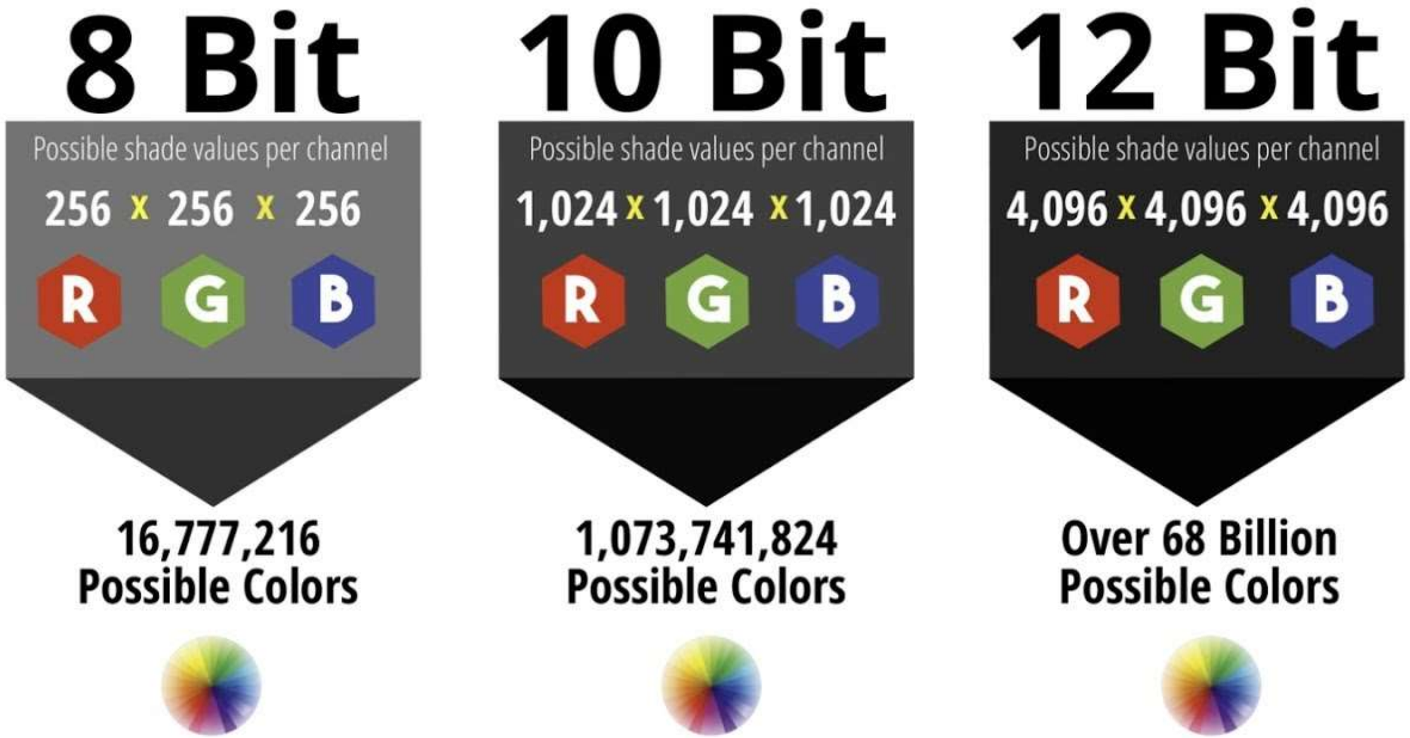
- **High Dynamic Rage (HDR)**
  - 정의
    - 넓은 dynamic range를 가지는 이미지
    - 픽셀 값이 luminance와 선형 관계 (근사)
  - 핵심 특징
    - pixel value ∝ 물리적 밝기
    - → photometric quantity 로 근사

LDR & HDR
Pixel Representation & Bit Depth

- **Low Dynamic Rage (LDR) Image**
  - 저장 방식
    - 보통 정수 (integer) 기반
    - 예: 8-bit / channel
    - 0~255 (총 256 단계)
  - 특징
    - 표현 가능한 값이 제한적
    - 밝기 범위가 좁음
    - 대부분 PNG, JPEG 등으로 저장
  - 결과
    - 밝은 영역, 어두운 영역 → clipping
    - → detail 손실

- **High Dynamic Rage (HDR)**
  - 저장 방식
    - 고비트 정밀도 (다양한 정의)
    - 10, 12, 16 bit integer
    - 32 bit floating point
  - 특징
    - 훨씬 넓은 값의 범위 표현 가능
    - 매우 밝은 값 / 어두운 값 모두 표현 가능
  - 결과
    - 장면의 실제 밝기 분포를 더 잘 보존

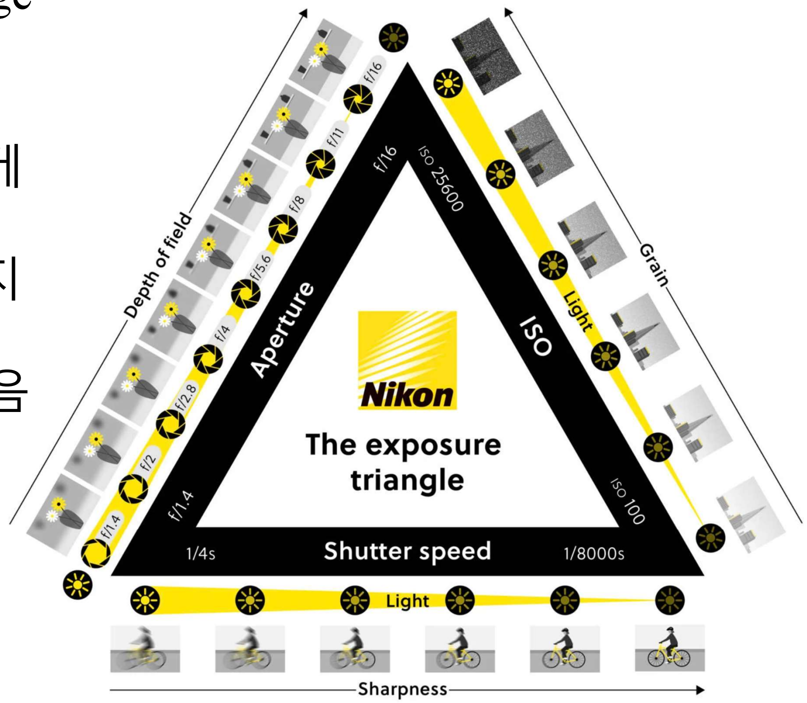
LDR & HDR
Pixel Representation & Bit Depth



- https://www.youtube.com/watch?v=6yXYxp0UiVg

HDR Imaging (HDRI, HDR Reconstruction)

- Multi / Single LDR 이미지로부터 High Dynamic Range Image 를 얻는 기법
  - 역광일 때, 배경은 밝고 사물이 어둡게 나오는 이미지
  - 밤에는 어두워서 밝기 노출 시간 (exposure time)을 길게 찍으면 블러가 생기고
  - 노출 시간을 짧게 하면 어두워서 원하는 이미지를 얻지 못하기 때문에
  - 일반 카메라, 블랙박스, cctv 등에서 많이 사용되고 있음



- https://www.nikon.com.cy/en_CY/learn-and-explore/magazine/tips-and-tricks/understanding-the-exposure-triangle

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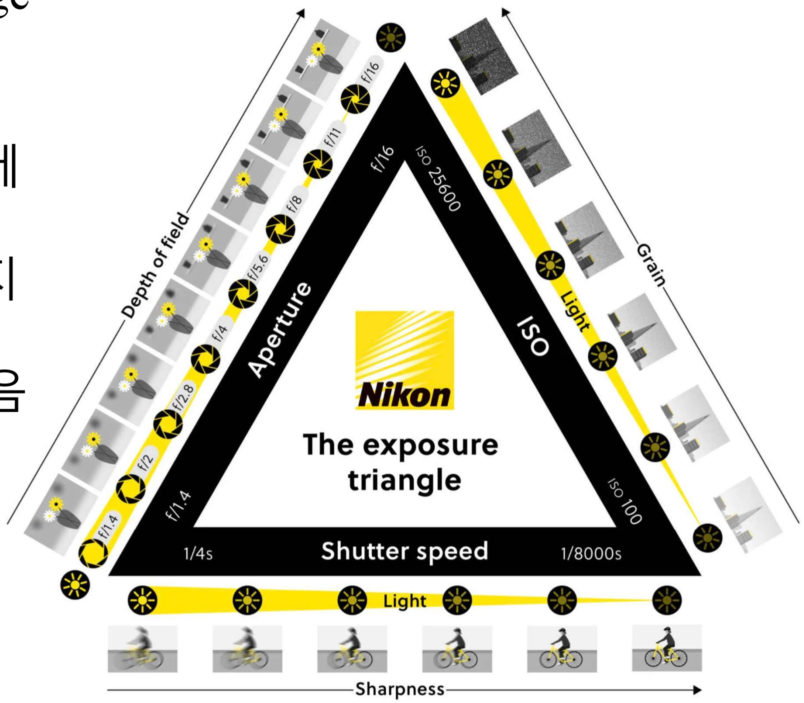

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Paper List | High Dynamic Range Imaging

This repository compiles a list of papers related to HDR. (Updating)

Continual improvements are being made to this repository. If you come across any relevant papers that should be included, please don't hesitate to open an issue.

Contents

Multi-Frame HDRI

Title	Paper	Code	Dataset	Key Words
AFUNet: Cross-Iterative Alignment-Fusion Synergy for HDR Reconstruction via Deep Unfolding Paradigm	ICCV-2025	AFUNet		
From Dynamic to Static: Stepwisely Generate HDR Image for Ghost Removal	TCSVT-2025			
UltraFusion: Ultra High Dynamic Imaging using Exposure Fusion	CVPR-2025	UltraFusion		
SAFNet: Selective Alignment Fusion Network for Efficient HDR Imaging	ECCV-2024	SAFNet		
PASTA: Towards Flexible and Efficient HDR Imaging Via Progressively Aggregated Spatio-Temporal Alignment	arxiv-2024			
Enhancing Multi-				

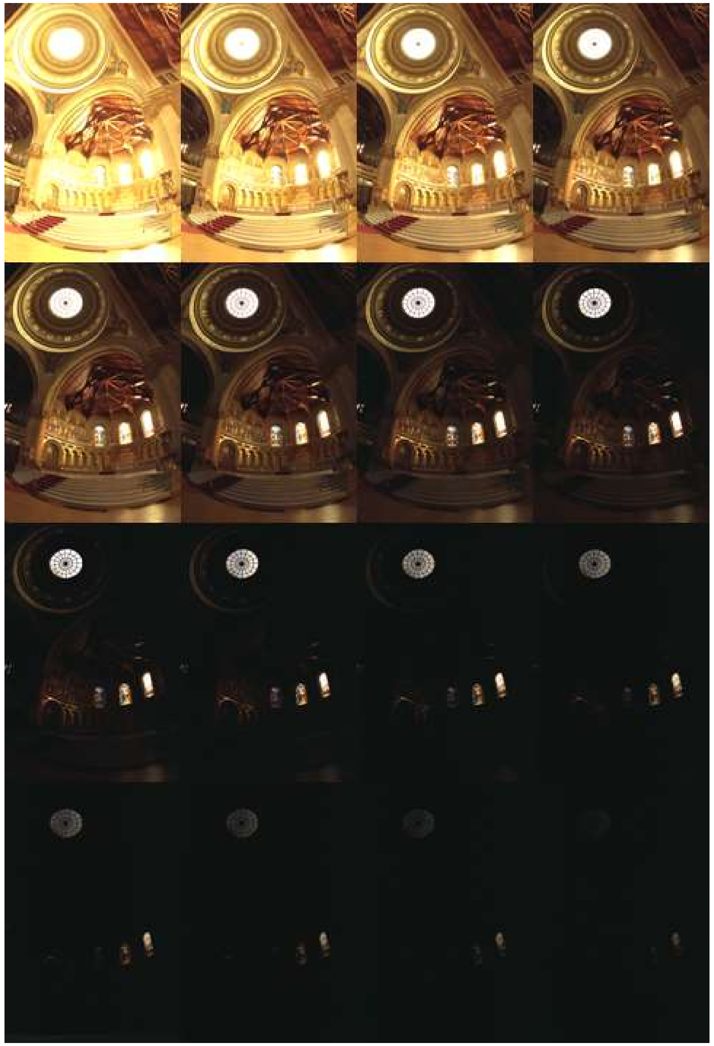
Single-Frame HDRI

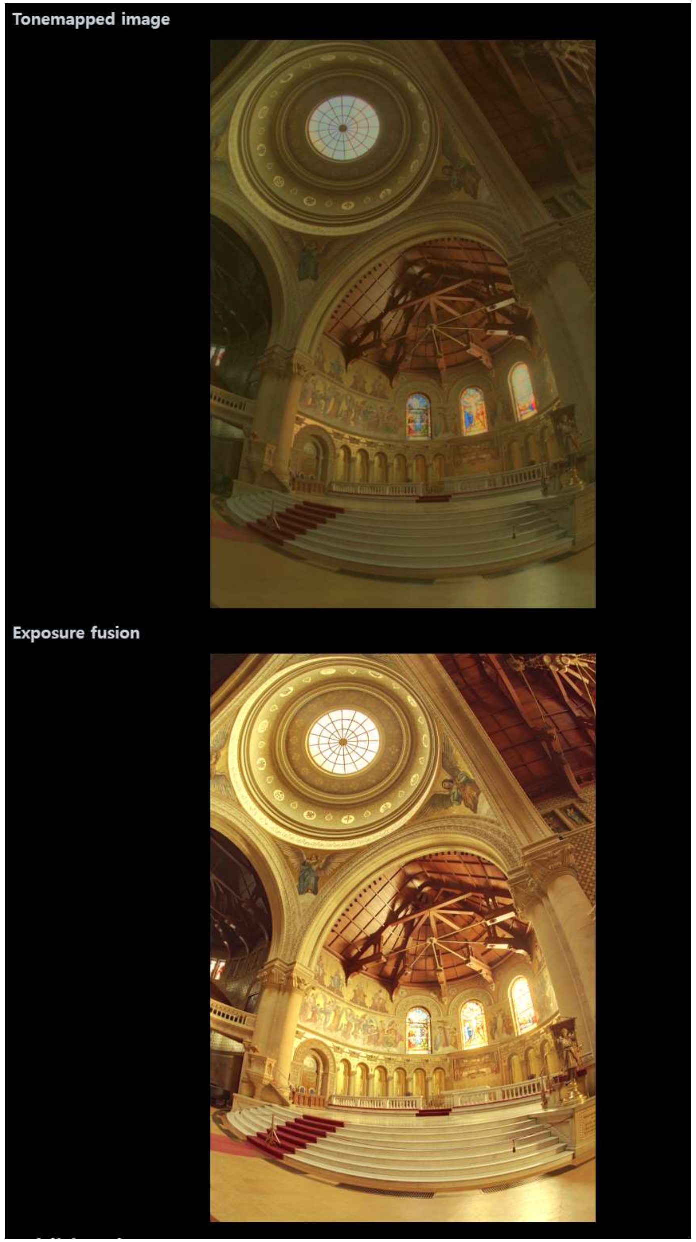
Title	Paper	Code	Dataset	Key Words
LEDiff: Latent Exposure Diffusion for HDR Generation	CVPR-2025	LEDiff		
DCDR-UNet: Deformable Convolution Based Detail Restoration via U-shape Network for Single Image HDR Reconstruction	CVPRW-2024			
Single-Image HDR Reconstruction by Multi-Exposure Generation	WACV-2023	code	DrTMO	
Single image ldr to hdr conversion using conditional diffusion	ICIP-2023			
RawHDR: High Dynamic Range Image Reconstruction from a Single Raw Image	ICCV-2023		RawHDR	
LHDR: HDR Reconstruction for Legacy Content using a Lightweight DNN	ACCV-2022	/		
Ultra-High-Definition Image HDR Reconstruction via Collaborative Bilateral Learning	ICCV-2021			
A Two-stage Deep Network for High Dynamic Range Image Reconstruction	CVPRW-2021	code	NTIRE 2021	

- https://github.com/rebeccaeexu/Awesome-High-Dynamic-Range-Imaging

Multi-Exposure Techniques

- Paul E Debevec and Jitendra Malik. Recovering high dynamic range radiance maps from photographs. In ACM SIGGRAPH 2008 classes, page 31. ACM, 2008.
- Mark A Robertson, Sean Borman, and Robert L Stevenson. Dynamic range improvement through multiple exposures. In Image Processing, 1999. ICIP 99. Proceedings. 1999 International Conference on, volume 3, pages 159–163. IEEE, 1999.
- Tom Mertens, Jan Kautz, and Frank Van Reeth. Exposure fusion. In Computer Graphics and Applications, 2007. PG'07. 15th Pacific Conference on, pages 382–390. IEEE, 2007.





- https://docs.opencv.org/4.x/d3/db7/tutorial_hdr_imaging.html.

Multi-Exposure Techniques

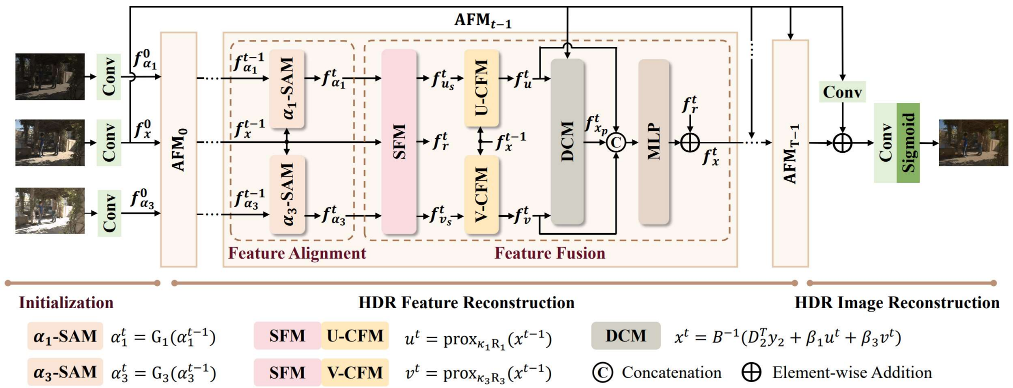


Figure 2. Framework of the AFUNet for HDR reconstruction consists of three main processes: Initialization, HDR Feature Reconstruction, and HDR Image Reconstruction. Within HDR Feature Reconstruction, stacked Alignment Fusion Modules (AFMs) iteratively refine target HDR features via alternating alignment and fusion. In the Feature Alignment subprocess, $f_{\alpha_1}^{t-1}$ and $f_{\alpha_3}^{t-1}$ are aligned with f_x^{t-1} by the Spatial Alignment Modules—denoted as α_1-SAM and α_3-SAM. In the Feature Fusion subprocess, the Spatial Fusion Module (SFM) performs preliminary optimization to obtain $f_{u_s}^t$ and $f_{v_s}^t$, followed by the Channel Fusion Modules—denoted as U-CFM and V-CFM—obtain f_u^t and f_v^t, respectively. Finally, the Data Consistency Module (DCM) obtains $f_{x_p}^t$, and an MLP with residual addition further refines f_x^t before it is transmitted to the next stage. The $\kappa_i = \frac{\lambda_i}{\beta_i}$, $\mathrm{R}_i = \mathrm{p}_i(\cdot, \alpha_i^t)$ $(i = 1, 3)$, and $B^{-1} = (D_2^{\mathrm{T}} D_2 + (\beta_1 + \beta_3) I)^{-1}$.

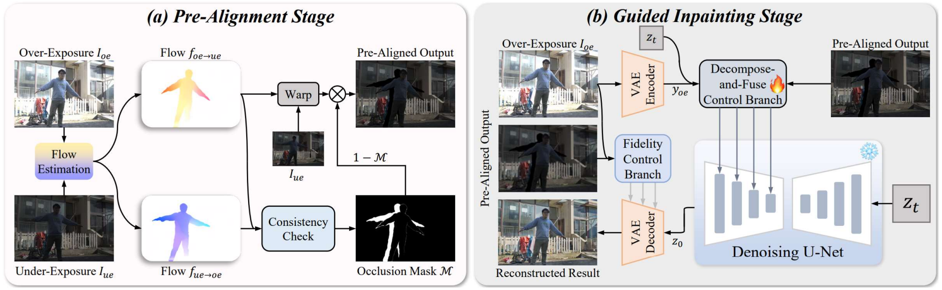


Figure 3. The whole backbone of UltraFusion. Our method is a 2-stage framework, consisting of (a) pre-alignment stage and (b) guided inpainting stage. The first stage pre-aligns the under-exposed image I_{ue} to the over-exposed image I_{oe} and masks the occluded regions. In the subsequent guided inpainting stage, we propose a new decompose-and-fuse control branch to utilize the diffusion priors, together with a fidelity control branch for fidelity keeping.

- AFUNet: Cross-Iterative Alignment-Fusion Synergy for HDR Reconstruction via Deep Unfolding Paradigm, ICCV 2025
- UltraFusion: Ultra High Dynamic Imaging using Exposure Fusion, CVPR 2025

LEDiff: Latent Exposure Diffusion for HDR Generation

Introduction

- Problem Statement
  - 현재 생성 모델들은 대부분 8-bit LDR image 생성
  - 문제:
    - highlight / shadow clipping
    - 실제 장면의 빛과 색 변화 표현 불가
    - tone mapping으로도 디테일 복원 불가능
  - Why this matters
    - LDR은 다음 작업에서 한계 발생:
      - image-based lighting, depth of field rendering
    - 기술 발전으로 HDR 지원하는 디스플레이 장치들 증가
      - 하지만 생성형 모델의 결과물(컨텐츠)들은 여전히 LDR

Introduction

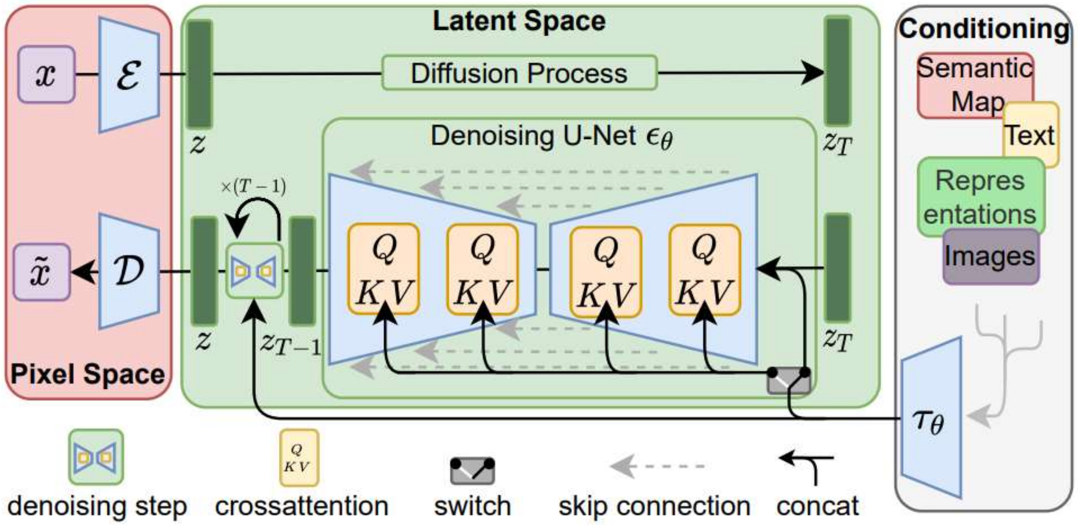
- Goal
  - Pretrained diffusion model을 HDR 생성 가능하게 확장
  - 조건:
    - 기존 생성 능력 유지
    - 소량 HDR 데이터 사용
- Method Summary
  - latent space에서:
    - highlight / shadow 복원
    - missing detail hallucination
  - decoder:
    - linearization
    - dynamic range 확장

Introduction

- Contribution
  - 1. Efficient HDR Fine-tuning
    - pretrained latent space 유지
    - 소량 HDR 데이터로:
      - dynamic range 확장
      - highlight / shadow 디테일 복원
  - 2. Generative HDR Framework
    - 기존 diffusion 모델 → HDR 생성 가능
    - LDR → HDR 변환에서도 SOTA 성능

Preliminaries : Latent Diffusion Model (LDM)

- LDM (a.k.a Stable Diffusion)
  - VAE Encoder E : 이미지 → latent로 압축
  - Denoising U-Net (Diffusion)
  - VAE Decoder : latent → 이미지로 복원



- Rombach, Robin, et al. "High-resolution image synthesis with latent diffusion models." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022.

Preliminaries : 왜 LDM은 HDR을 못 만드냐

- latent 자체가 이미 "LDR 공간"이다
  - VAE가 LDR 데이터로만 학습되어 있음
  - 이에 따라 denoiser도 LDR latent만 생성하도록 학습되어 있음

- naive approach
  - VAE + Unet 모두 HDR로 finetune, training
  - Too expensive
  - pretrained 모델의 generative prior 손상

- latent space 그대로 유지한다!

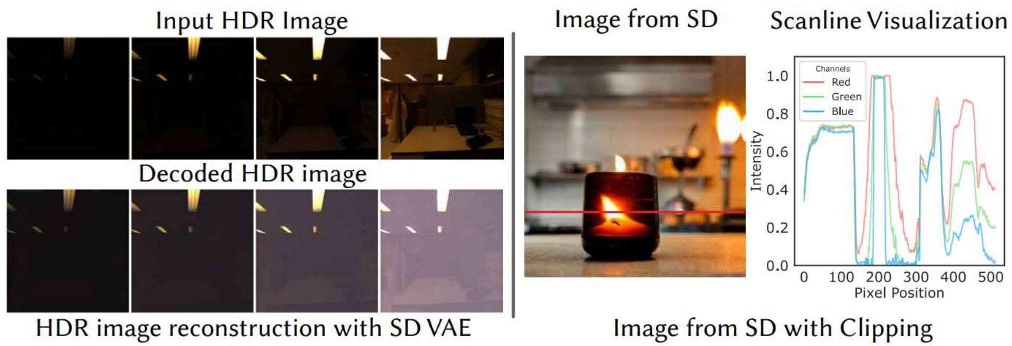


Figure 3. Limitations of the vanilla Stable Diffusion (SD) in generating HDR content. **Left:** The limitation of the SD VAE in encoding and decoding an HDR image, visualized in multiple exposure levels, which reveals a significant fidelity loss, especially in the shadow. **Right:** An image generated with SD, along with a scanline that shows pixel clipping in highlight and shadow regions.

Method

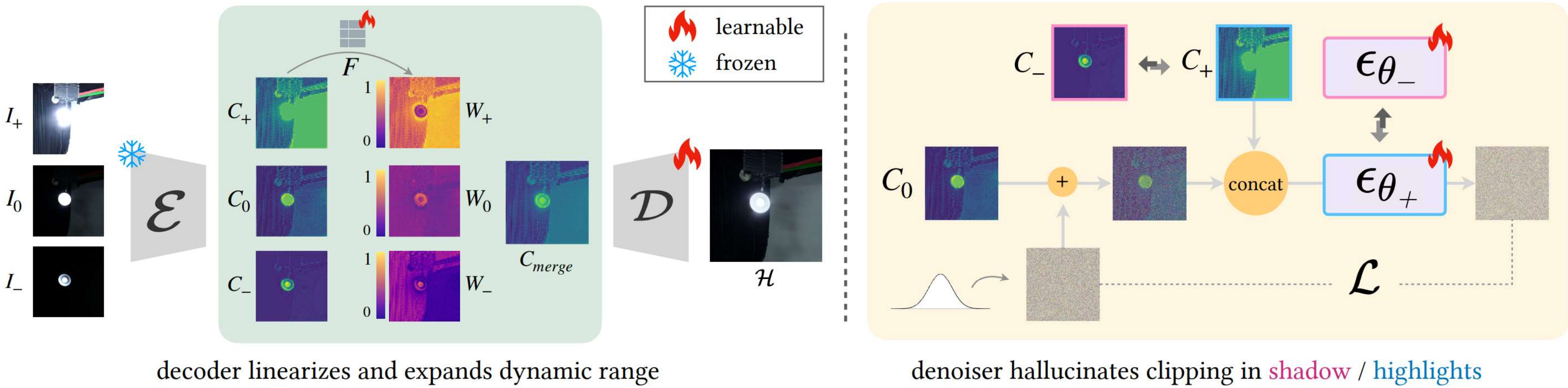


Figure 2. Finetuning scheme of decoder and denoiser. (Left: fine-tuning the decoder) Exposure bracketed images I_+, I_0, I_- are encoded via the pre-trained encoder to generate corresponding latent codes. These latent codes are fused using a learnable fusion module $\mathcal{F}$ to produce a latent code $\mathcal{C}_{\text{merge}}$ free of clipping, which is then decoded into an HDR image $\mathcal{H}$ through the finetuned decoder. (Right: fine-tuning the denoiser) The model takes as input the latent code $\mathcal{C}_+$ for training highlight denoiser ϵ_{θ_-} or $\mathcal{C}_-$ for training shadow denoiser ϵ_{θ_+}, along with a $\mathcal{C}_0$ corrupted by randomly sampled noise.

Merging Latent Exposure Brackets

- HDR image $H \rightarrow$ 여러 LDR image 생성 I_{-}, I_0, I_{+}
- Pretrained VAE Encdoer E
  - $C_{-},\ C_0, C_{+} = E(I_{-}), E(I_0), E(I_{+})$
- Training learnable module F
  - $C_{merge} = F(C_{-},\ C_0, C_{+})$
- Fine-tuining VAE Decoder D
  - $\widehat{H} = D(C_{merge})$

$$\mathcal{L}_{\mathcal{F},\mathcal{D}} = \mathcal{L}_{\text{rec}} + \lambda\mathcal{L}_{\text{GAN}} \tag{2}$$

Generating Latent Exposure Brackets

- 단일 LDR latent code C가 주어졌을 때, 이를 exposure bracket {C−,C0, C+}로 확장
  - Text-to-HDR generation
  - LDR-to-HDR conversion
- Highlight hallucination denoiser $\epsilon_{\theta\_}$
  - Fine-tuning LDM denoiser
- Shadow hallucination denoiser ϵ_{θ_+}
  - Fine-tuning LDM denoiser

uses a standard denoising diffusion loss.

To train $\epsilon_{\theta\_}(\cdot)$, we first randomly sample an exposure bracket from the HDR dataset, following the process outlined in Sec 3.2. These samples are encoded using the pre-trained VAE encoder, producing a latent exposure bracket $\{C_-, C_0, C_+\}$. We then corrupt the underexposed latent C_- and C_0 with Gaussian noise at a randomly sampled timestamp t. The task of the denoising network, $\epsilon_{\theta\_}(\cdot)$ is to predict the noise $\hat{\epsilon}_{C_-,C_0} = \epsilon_{\theta\_}(C_0, C_-^t, t)$ and $\hat{\epsilon}_{C_0,C_+} = \epsilon_{\theta\_}(C_+, C_0^t, t)$ using the objective:

$$\mathcal{L}_C = \mathbb{E}_{C,\epsilon_{C_-,C_0} \sim \mathcal{N}(0,\mathcal{I}), t\sim\mathcal{U}(T)} \left\| \epsilon_{C_-,C_0} - \hat{\epsilon}_{C_-,C_0} \right\|_2^2 \quad (3)$$

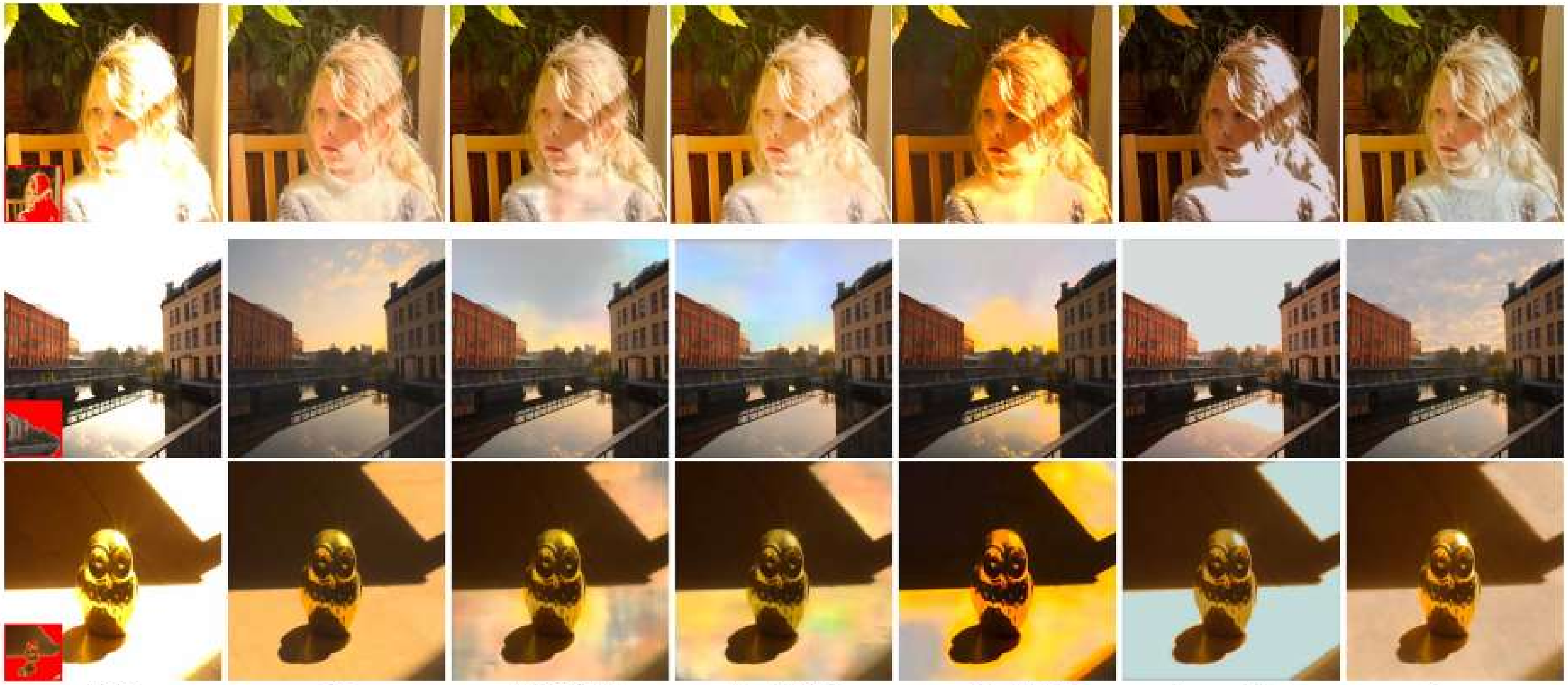
$$+ \mathbb{E}_{C,\epsilon_{C_0,C_+} \sim \mathcal{N}(0,\mathcal{I}), t\sim\mathcal{U}(T)} \left\| \epsilon_{C_0,C_+} - \hat{\epsilon}_{C_0,C_+} \right\|_2^2 \quad (4)$$

During inference, given an LDR image, we iteratively apply the fine-tuned denoiser to generate its corresponding lower-exposure latent codes.

For generating a higher-exposure latent code C_+, C_0 that hallucinates shadow details, we follow the same procedure to fine-tune a conditional denoiser $\epsilon_{\theta_+}(\cdot)$ with parameters θ_+, conditioned on C_0 and C_- respectively.

After generating the latent exposure bracket, we merge it using $\mathcal{F}$, then decode the merged latent code into an HDR image with the fine-tuned HDR decoder.

Highlight Hallucination Comparison with Generalized HDR Reconstruction Methods



LDR GT HDRCNN MaskHDR SingleHDR ExpandNet Ours

Figure 5. LDR-to-HDR image reconstruction comparisons. Our method effectively hallucinates details in both over- and under-exposed

Method	Highlights					Shadow				
	HDR-VDP3 ↑	PU21-PIQE ↓	FID-R ↓	FID-D↓	FID-L↓	HDR-VDP3 ↑	PU21-PIQE ↓	FID-R ↓	FID-D↓	FID-L↓
HDRCNN [20]	6.90 ± 1.10	49.43 ± 6.91	13.39	16.95	16.67	6.56 ± 1.10	70.95 ± 7.84	32.55	44.20	26.41
MaskHDR [56]	6.47 ± 1.06	49.38 ± 6.95	12.83	13.85	15.21	6.49 ± 1.24	70.98 ± 8.64	32.54	44.43	26.10
SingleHDR [38]	6.13 ± 0.87	49.74 ± 7.29	28.68	34.53	29.50	7.47 ± 0.78	46.04 ± 9.41	39.99	57.28	37.17
ExpandNet [42]	5.23 ± 1.61	52.53 ± 6.50	18.85	25.49	21.34	5.40 ± 0.93	60.01 ± 8.37	32.60	45.50	29.10
Ours	6.16 ± 0.97	48.46 ± 7.04	12.70	13.08	13.73	6.98 ± 0.83	43.37 ± 8.04	20.93	25.54	24.26
GlowGAN [63]	5.67 ± 1.24	46.18 ± 8.22	27.44	40.43	43.85	–	–	–	–	–
Ours∗	$\underline{5.92 \pm 1.17}$	$\underline{45.29 \pm 7.27}$	$\underline{15.62}$	$\underline{17.89}$	$\underline{16.19}$	–	–	–	–	–

Table 1. Quantitative evaluation on content hallucination (Ours∗ denotes the test results of our method on a subset of the test set, selected to meet GlowGAN's requirement for class-specific inputs). Full-reference metrics are less suited to generative models due to natural deviations in hallucinated content from the original HDR scene. Our method achieves competitive performance on the full-reference HDR-VDP3 metric and outperforms alternative methods on the no-reference PU21-PIQE metric, which better captures image naturalness and perceptual fidelity. To complement these metrics, we use FID [28] to evaluate the distribution of generated and ground-truth HDR images. Since FID is optimized for LDR, we tone map all HDR images using three tone mapping operators, resulting in three FID scores: FID-R [53], FID-D [19], and FID-L [37]. Across all tone-mapping methods, our method consistently achieves lower FID scores, demonstrating closer alignment with the HDR ground-truth distribution and superior overall quality.

- https://www.hdrsoft.com/resources/dri.html#hdri
- https://www.anyhere.com/gward/hdrenc/hdr_encodings.html
- https://openexr.com/en/latest/TechnicalIntroduction.html#the-half-data-type
- https://www.cgkorea.co.kr/global/library/The_Ins_and_Outs_of_HDR_What_is_HDR.html
- https://doinghun.com/image-signal-processing-isp/
- https://news.samsungdisplay.com/14820
- https://en.wikipedia.org/wiki/High_dynamic_range#High-dynamic-range_imaging
- https://docs.opencv.org/3.4/d2/df0/tutorial_py_hdr.html
- https://www.cl.cam.ac.uk/~rkm38/pdfs/mantiuk15hdri.pdf